



TIPS & TRICKS

Creating a partition larger than 3TB in Linux

Many of us do not know that we cannot create a Linux partition larger than 2TB using the *fdisk* command. Using *fdisk* is fine for desktop and laptop users, but not when it comes to servers and especially a storage device, where you need a large partition. For example, you cannot create a 3TB or 4TB partition size (RAID based) using the *fdisk* command while building a NAS solution.

To solve this problem, you can use the GNU *parted* command with GPT. It supports Intel EFI/GPT partition tables.

Here is an example of how we can use *parted*. We can use the following command to find the exact size of the HDD:

```
# fdisk -l /dev/sdb
```

The output will show the exact disk size and the partition details. In our case, the disk size is 3000.6GB.

So we will be using *parted* to create a partition instead of *fdisk*, as follows:

```
# parted /dev/sdb
```

```
GNU Parted 2.3
Using /dev/sdb
Welcome to GNU Parted! Type 'help' to view a list of
commands.
(parted)
```

Now create a new GPT disklabel:

```
(parted) mklabel gpt
```

The output will be as shown below:

```
Warning: The existing disk label on /dev/sdb will be
destroyed and all data on this disk will be lost. Do you
want to continue?
Yes/No? yes
(parted)
Next, set the default unit to TB, enter:
```

```
(parted) unit TB
```

To create a 3TB partition size, enter:

```
(parted) mkpart primary 0 0
```

Or:

```
(parted) mkpart primary 0.00TB 3.00TB
```

To print the current partitions, enter:

```
(parted) print
```

The output will be:

```
Model: ATA ST33000651AS (scsi)
Disk /dev/sdb: 3.00TB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Number Start End Size File system Name Flags
1 0.00TB 3.00TB 3.00TB ext4 primary
```

Quit and save the changes, enter:

```
(parted) quit
```



Note: You may need to update */etc/fstab*.

Once you are done with the partitioning of the disk, you need to format and create a file system to make the disk useable. Use the *mkfs.ext3* or *mkfs.ext4* commands to format the file system, by entering the following command:

```
# mkfs.ext3 /dev/sdb1
```

Or:

```
# mkfs.ext4 /dev/sdb1
```

Once the formatting is complete, you have to mount the partition, by using the following commands: